

ARAIN, TARIQ ALI

Doctor of Philosophy

@ tariqaliarain@hhu.edu.cn @ tariqkavisharain@yahoo.com WeChat ID: kavish_ali +86 139 5100 3958
LinkedIn Google Scholar 13951003958 Nanjing, CN



RESEARCH INTERESTS

Deep Learning | Computer Vision | Object Detection | Person Re-Identification

EDUCATION

Doctor of Philosophy (PhD) | College of Computer Science and Software Engineering

Hohai University (HHU)

Sep 2021 – March 2026 Nanjing, China

- Dissertation: Advanced Deep Learning Techniques for Person Re-Identification in Surveillance Systems.
- Research Focus: computer vision, person re-identification (ReID), low-resolution and occlusion-robust learning, attention-based transformers, deep metric learning, cross-resolution feature alignment.
- Technical Expertise: PyTorch, Python, CUDA-based GPU acceleration, distributed training (DDP), TorchScript, large-scale dataset management, visualization, and model interpretability.
- Research Output: multiple peer-reviewed publications, with several first-author manuscripts currently under second-round review.

Master of Engineering | School of Computer Science and Engineering

Nanjing University of Science and Technology (NJUST)

Sep 2017 – March 2022 Nanjing, China

- Thesis: Research on the Optimization of Ticket Assignment Problem in Large Data Centers.
- Research Areas: multi-objective optimization, heuristic and meta-heuristic algorithms, scheduling, and large-scale resource allocation.
- Methods: algorithm design, complexity analysis, simulation, and performance evaluation.

Bachelor of Science | Bachelor of Information Science and Technology

Shah Abdul Latif University (SALU)

Apr 2009 – June 2013 Khairpur Mirs, Pakistan

- Final Year Project: Developed a Library Management System using Visual Basic (VB) and SQL Server.

PUBLICATIONS FIND HERE

- Swin Transformer with Attention Mechanism: A Novel Framework for Person Re-Identification. Tariq Ali ARAIN, Pengcheng Zhang, Qing Meng*, Abdullahi Uwaisu Muhammad. Pattern Analysis and Applications (SCI), Vol. 28, Article 95, 2025. Springer Nature. DOI: [🔗](#)
- An Adaptive Cross-Resolution Network for Low-Resolution Person Re-Identification. Tariq Ali ARAIN, Pengcheng Zhang, Sehrish Mazhar, Qing Meng*. International Journal of Pattern Recognition and Artificial Intelligence (SCI). DOI: [🔗](#)
- FocusReID: Densely Connected Fusion Attention Networks for Real-World Person Re-Identification. Tariq Ali ARAIN, Pengcheng Zhang, Sehrish Mazhar, Qing Meng* and Imran Ali Soomro. Image and Vision Computing (SCI). DOI: [🔗](#)

SUMMARY

Ph.D. in Software Engineering from Hohai University (HHU), completed in March 2026, supervised by Prof. Pengcheng Zhang (Google Scholar). My research focuses on deep learning, computer vision, and robust AI systems, with particular expertise in person re-identification, object recognition, transformer-based architectures, metric learning, and deployment-oriented visual perception. I am especially interested in applying artificial intelligence and data-driven methods to real-world industrial challenges, including autonomous-driving perception, intelligent mobility, surveillance, and safety-critical visual perception. My future work aims to advance multimodal and vision-language learning, efficient model design, and interpretable AI systems for reliable deployment in complex real-world environments.

SKILLS

Deep Learning & Computer Vision: Python, PyTorch, OpenCV, NumPy, Pandas, CUDA, MATLAB

Other Programming Languages: HTML, CSS, R, Java, C, C++

Software & Tools: PyCharm, Visual Studio, Git/Git Bash, Docker, Jupyter, Microsoft Office, LaTeX, Adobe Photoshop, VirtualBox

Operating Systems: Windows, Linux (CentOS, Ubuntu, etc.)

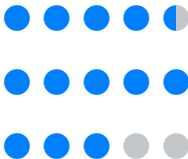
Vision & ML: representation & metric learning, transformer attention, domain adaptation, interpretability (Grad-CAM), real-world vision.

Languages:

English
Professional Proficiency

Urdu
Native Speaker

Chinese
HSK Level 3



CERTIFICATIONS



Build Generative Adversarial Networks
Coursera | DeepLearning.AI (Feb 2023).

- *Multi-objective Optimization of Ticket Assignment Problem in Large Data Centers.*
Tariq Ali Arain, Xiangjie Huang, Zhicheng Cai*, Jian Xu. *CCF Conference on Computer Supported Cooperative Work and Social Computing 2021 (CCF B)*., Vol. 1492. Springer Nature. DOI: [🔗](#)
- *A Novel Multi-Objective Optimization Approach for Ticket Assignment in Large Data Centers.*
Tariq Ali Arain*, Sehrish Mazhar. *Cluster Computing (SCI)*. **Accepted on 6/6/2026. in Cluster Computing**
- *Optimizing Service Placement in Fog Environments with Deep Reinforcement Learning.*
Wei Liu, Qin Xie, Tariq Ali Arain, Usama Ali. *ICSS 2025. Lecture Notes in Computer Science*, pp. 288–302. Springer Nature. DOI: [🔗](#)
- *Efficient Workload Prediction Model for Containerized Cloud.*
Tianqi Xie, Yunfei Zhang, Changhong Tai, Tariq Ali Arain, Usama Ali. *ICSS 2025. Lecture Notes in Computer Science*, pp. 303–315. Springer Nature. DOI: [🔗](#)
- *FedQR: Communication-Efficient Federated Learning via QR Factorization.*
Abdoul Fatakhou Ba, Yingchi Mao, Hamza Djigal, Abdullahi Uwaisu Muhammad, Tariq Ali Arain, Siaka Konate. *ICONIP 2024. Lecture Notes in Computer Science*, pp. 433–448. Springer Nature. DOI: [🔗](#)

WORK EXPERIENCE

AI Research Engineer

Hohai University

📅 Sep 2021 – March 2026

📍 Nanjing, China

- Conducted research in **Person Re-Identification (ReID)** for real-world surveillance, focusing on low-resolution, occlusion, and cross-camera variation; developed *SwinReID*, *FocusReID*, and *VarReID* using transformers, attention, multi-scale fusion, and metric learning.
- Designed efficient deep-learning pipelines using PyTorch, AMP, DDP multi-GPU training, pruning/compression, and inference acceleration.
- Developed research methods in federated learning, edge/cloud optimization, and workload forecasting, including *FedQR*, *HA3C*, and *Flash-Attention + LSTM* models. [🔗](#) [🔗](#) [🔗](#)
- Applied robust visual perception techniques to deployment-oriented AI problems, including occlusion handling, domain shift analysis, low-resolution recognition, and efficient model design, with relevance to autonomous-driving perception, intelligent surveillance, and industrial AI systems.

TEACHING EXPERIENCE

Computer Instructor

Sabheeta Academy of Computer Science and Technology

📅 Aug 2012 – Jul 2013

📍 Khairpur, Pakistan

- Taught foundational computer science including programming, office software, and hardware basics

Trainee

VifTech Solution

📅 May 2016 – Apr 2017

📍 Karachi, Pakistan

- Learned basic knowledge of HTML, CSS, Bootstrap, JavaScript, jQuery, Object-Oriented Programming (OOP) with .NET, C#, and Database management
- Collaborated in designing and implementing project plans and schedules

VOLUNTEER EXPERIENCE

Public Welfare & Cultural Activities, Nanjing, China Supported community safety initiatives ("Harmony Month") and traditional cultural events at Lingxing Academy (Confucian Temple). International Culture Festival, Hohai University Coordinated festival activities and intercultural engagement.



Advanced Machine Learning (ML)
Coursera|Stanford Online (Dec 2022).



Supervised Machine Learning
Coursera|Stanford Online (Nov 2022).



Database Computer Training Course
Pro-Logics (Dec 2016 – Mar 2017).



C# Programming Course
Aptech Computer Education (2016).

ACADEMIC ACTIVITIES

- **CCF 18th International Conference on Service Science (ICSS 2025)**
Nanjing, China | May 9–11, 2025
- **IAHR Professional Workshop ("My Professional Adventure")**
Nanjing, China | Oct 19, 2023
- **SECAT Conference (Scientific Expertise, Communication and Trust)**
Aarhus, Denmark | Sep 27–28, 2022
- **Academic Training Program, School of Architecture, Southeast University**
Nanjing, China | Oct 10–24, 2022
- **IPMC – International Pure Mathematics Conference**
Islamabad, Pakistan | Aug 23–25, 2021
- **Peer Reviewer – International Journal of Pattern Recognition and Artificial Intelligence (SCI)**

ACADEMIC PROJECTS

- Contributed to a research project supported by the National Natural Science Foundation of China.
- Played a key role in a project funded by the China Postdoctoral Science Foundation.
- Contributed to a research project supported by the National Natural Science Foundation of China and the Fundamental Research Funds for the Central Universities.

AWARDS

- 🏆 Fully funded Master's scholarship (Nanjing Municipal Government), NJUST.
- 🏆 Fully funded PhD scholarship, Hohai University (HHU).

REFEREES

Voice Dean Prof. Pengcheng Zhang

@ pchzhang@hhu.edu.cn

✉ Hohai University, Nanjing 210098, Jiangsu, P. R. China

Key Laboratory of Water Big Data Technology of Ministry of Water Resources
College of Computer Science and Software Engineering

Prof. Dr. Yan Tang

@ tangyan@hhu.edu.cn

✉ Hohai University, Nanjing, P. R. China
College of Computer Science and Software Engineering